AIRFIELD PLANNING & WAIVERS PLAYBOOK

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Airfield Planning & Waivers Playbook

Introduction & Background

The intended audience of the *Airfield Planning & Waivers Playbook* is the Installation's Airfield Planning & Waivers Program Manager (AP&W PM) and team. This playbook addresses the method for correctly identifying obstructions, completing airfield waiver requests, reducing active waivers where applicable, and mitigating the risk posed by obstructions through programmed projects. Guided checklists are included at the end of each section to assist in the obstruction management process or when completing airfield waiver request packages and annual waiver review packages. For questions and concerns regarding the Airfield Planning & Waivers Program, please contact AFCEC/CP at AFCEC.CP.workflow1@us.af.mil.

The Airfield Planning & Waivers (AP&W) Program transitioned in April 2018 from AFIMSC/IZ to AFCEC/CP. AFCEC/CP serves as the technical review entity for all permanent and temporary Air Force waiver requests prior to their submittal for MAJCOM approval. *Airfield and Airspace Waiver Request Form*, AF Form 505 was created as part of the revitalization efforts for the program, mandated by AFI 32-1015, *Comprehensive Planning*. More Airfield Planning & Waiver documents and information is located in the Airfield Planning & Waivers Toolbox. Forms and guidance are simplified for ease of use and are updated regularly to reflect the latest changes in the AP&W Program.

Common Terminology

Airfield Planning. A comprehensive process to address airfield development, waivers to airfield criteria, focusing on investment strategy to reduce the number of temporary airfield waivers, eliminate unnecessary obstructions, and promote flying safety.

Airfield District. Airfield/ Flightline Districts encompass an airfield's runways and their associated taxiways, taxi lanes, parking ramps/aprons, aircraft hangars, Aircraft Maintenance Units (AMUs), squadron operations, aerospace ground equipment (AGE), back shops, the air traffic control tower (ATCT), and administrative facilities directly related to flight operations or aircraft maintenance. This district is largely industrial and utilitarian in nature, comprising facilities and land uses almost exclusively dedicated to the support of airfield operations.

Accident Potential Zones (APZs). Includes the Clear Zone (CZ), APZ I, and APZ II. Each of these zones is based upon historical data of aircraft accidents and include recommendations for what type of development should and should not occur within them. For example, it is recommended that no inhabitable development take place within a CZ because this is the most likely location for an aircraft accident to occur.

Additional terminologies with illustrating figures continue on next pages.

Building Restriction Lines (BRLs) are established laterally from the runway centerline along the predominant line and height of flight-line facilities at installation airfields per Unified Facilities Criteria (UFC) 3-260-01, Appendix B Section 17. BRLs set an elevation control line at the predominant roofline of facilities within the designated zone. Follow UFC BRL guidance to rationalize new development of flightline facilities within the exempted area. Future modification to Building Restriction Lines once they have been established is generally prohibited. AFCEC/CP will review BRL modification requests made by an installation on a case-by-case basis to ascertain whether the request is justified. (See Figure 1)

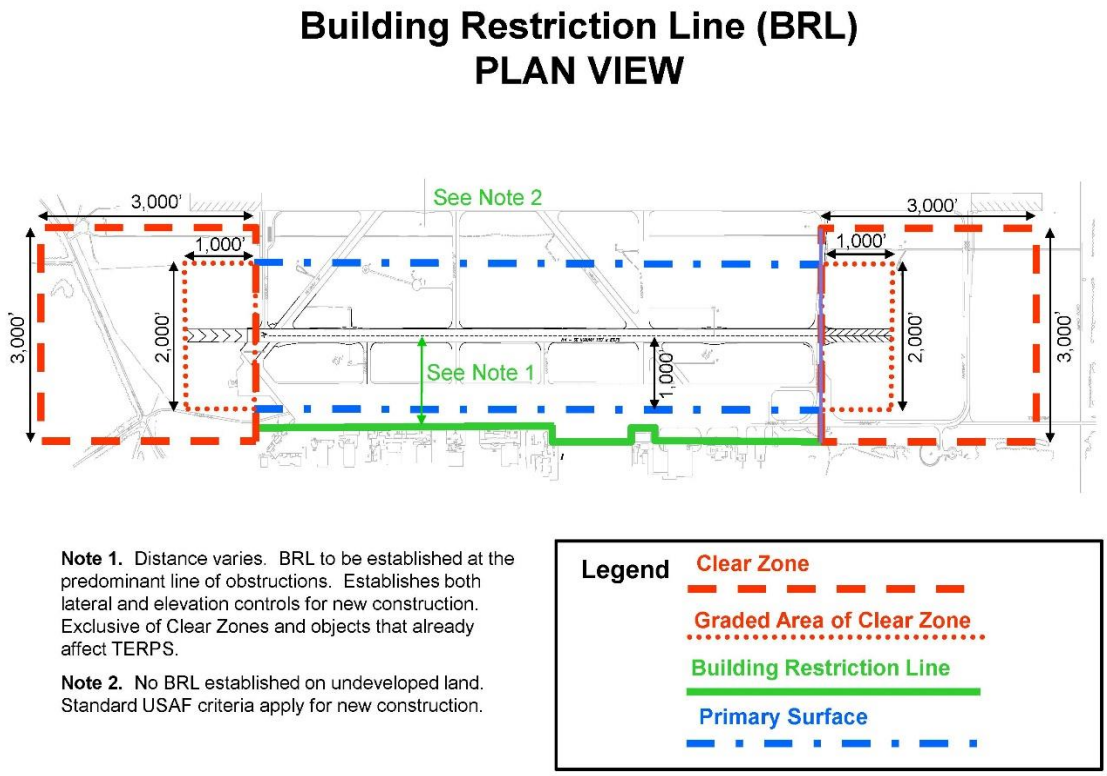

Figure 1 Building Restriction Lines (BRL), Plan View

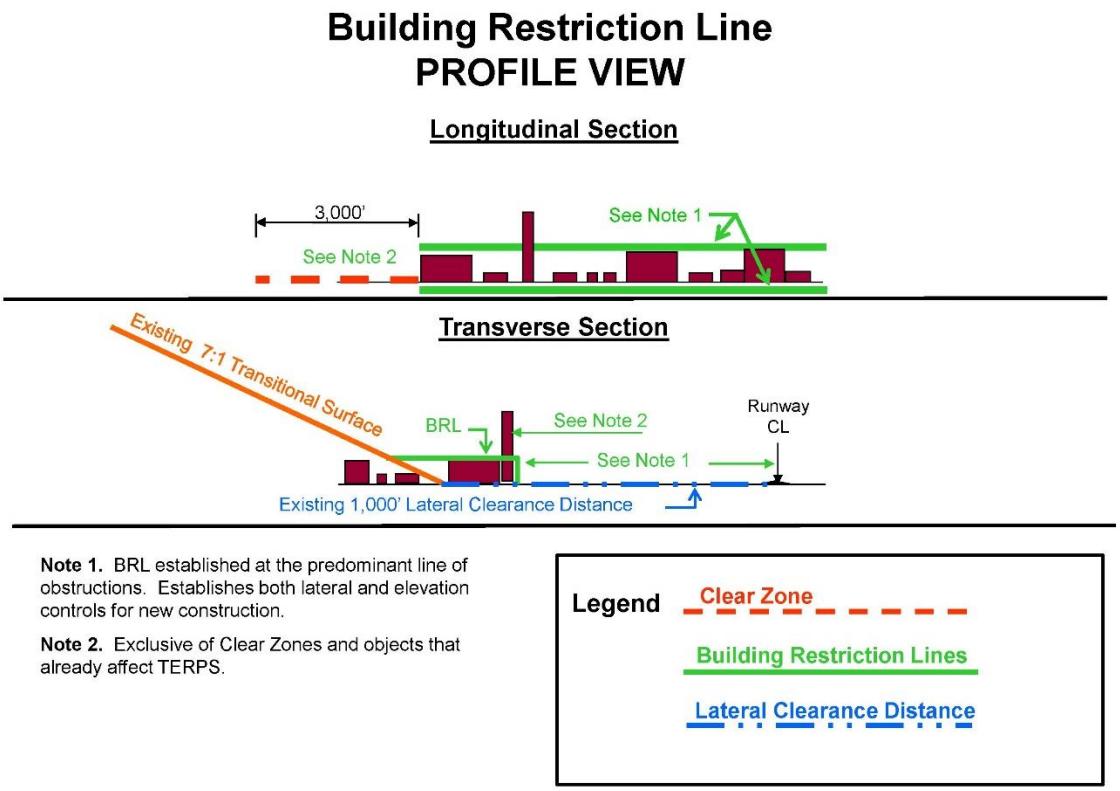

Figure 2 Building Restriction Lines (BRL), Profile View

Imaginary Airspace Surfaces. Imaginary surfaces are established in space around airfields in relation to runways, helipads, or helicopter runways, and are designed to define the obstacle free airspace around an airfield. (See Figure 2)

UFC 3-260-01 Chapter 3 defines seven types of imaginary surfaces: primary/lateral clearance zone, approach-departure clearance (ADCS, 50H:1V ratio), inner horizontal (150' elevation), conical (fixed-wing only, 20H:1V ratio), outer horizontal (fixed-wing only, 500' elevation), transitional (7H:1V ratio), and graded portion of the clear zone. Imaginary surfaces build upon each other and are designed to eliminate obstructions to air navigation and operations, either natural or man-made. Each type of imaginary surface has different dimensions and different planes or slopes. Structures protruding into a surface may be considered vertical obstructions. (See Figure 2)

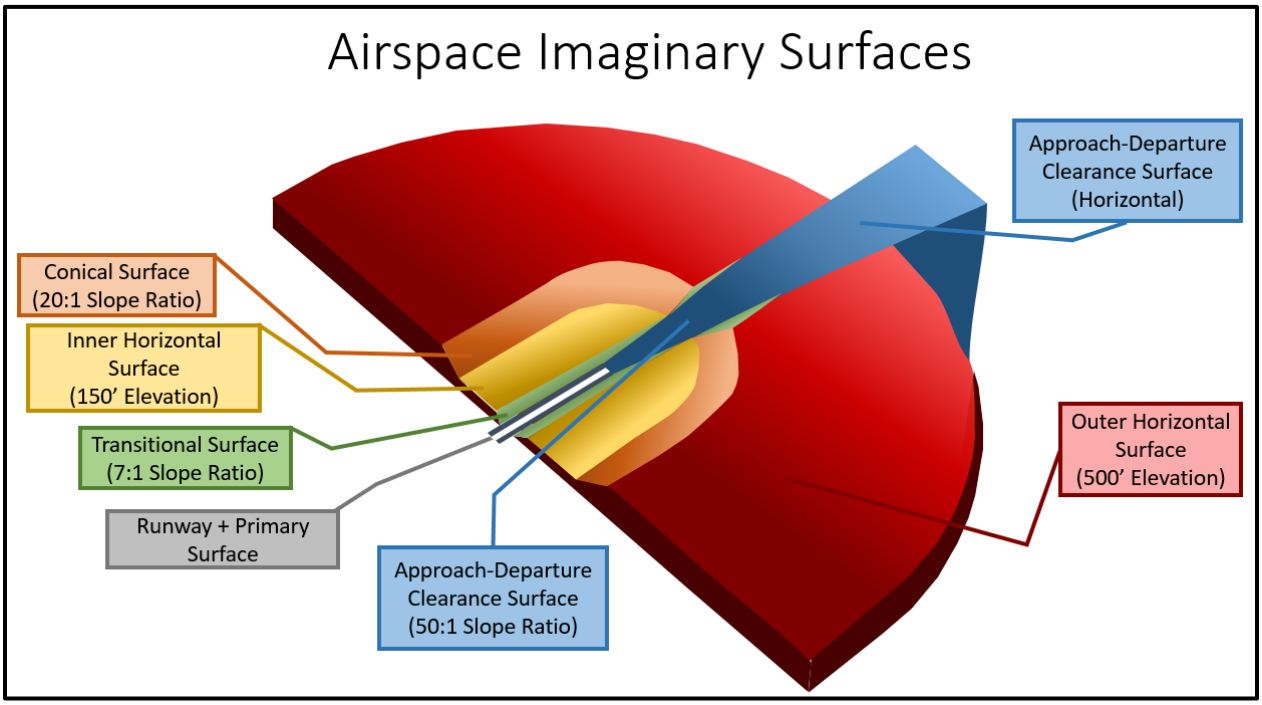

Figure 3 Airspace Imaginary Surfaces (Class B)

Obstruction Management

This section provides details on obstruction management tools and techniques for identifying and managing airfield obstructions.

What is an Obstruction?

An obstruction is a natural or man-made object that violates airfield or heliport clearances or projects into imaginary airspace surfaces.

Obstruction Identification & Airfield Assessment

At minimum, the installation AP&W Program Manager participates in an annual assessment of the airfield and airspace criteria in accordance with UFC 3-260-01 B1-2.5.4. Per AFI 13-204 Vol 2, *Airfield Operations Standardization and Evaluations*, the installation Airfield Manager has a requirement to conduct an annual airfield compliance/safety inspection. The AP&W Program Manager should take the opportunity to join this inspection and conduct the installation airfield assessment. If a potential obstruction is identified, an assessment must be conducted to verify any violations to airfield criteria. In instances where an obstruction violates airfield criteria, the installation AP&W Program Manager shall provide an evaluation after determining the correct course of action, whether that is to mitigate, correct, or remove the violation in question.

Once identified, obtain the following information to aid in accurately describing the obstruction:

- List of imaginary surfaces and UFC criteria violated by the obstruction
- Map of location(s) using ArcGIS or GEOBASE
- Photos of obstruction(s) and applicable measurements
- Project programmed to correct the violation (temporary waivers only)

Items listed above are used to create the airfield waiver request package.

The identification of an obstruction generally necessitates the request for a new permanent or temporary airfield waiver, depending on the nature of the violation. Permanent waivers are requested when violations cannot be reasonably corrected *and* pose little to no threat to flying operations, whereas temporary waivers are reserved for correctable violations. Both types of waivers require MAJCOM Deputy Commander approval after installation level coordination is complete.

In conjunction with the waiver request, an Obstruction Evaluation/Airport Airspace Analysis (OE/AAA) request may need to be submitted to the FAA using Form 7460-01, *Notice of Proposed Construction or Alteration* to ensure compliance with criteria. The FAA provides an evaluation and Part 77 Case/ASN number (*see the Airfield Waivers Section for reference to the complete airfield waiver request process*).

Obstructions must also be addressed during the installation's Annual Airfield and Airspace Waiver Review. Waivers are not required for exemptions or permissible deviations. However, waived obstructions, and exemptions, must be clearly identified in the waiver list, annual review meeting minutes and waiver map (*see the Annual Airfield and Airspace Waiver Review Section for the complete waiver review process*).

Airfield Management (AM) will issue NOTAMs, as required, to identify obstructions outlined in a temporary or permanent waiver and its associated maps. Appropriate NOTAMs should be issued after the approval of Temporary Construction or Air Show waivers.

Obstructions Outside of the Installation Boundary. UFC 03-260-01 and other airfield policy does not permit waivers for obstructions situated beyond the installation boundary. This includes, but is not limited to, naturally occurring topographic features such as mountains, trees, lakes, and canyons, or man-made obstacles such as wind turbines, cell-phone towers, overhead power transmission lines. Any existing waivers that address outside obstructions should be removed from the installation's portfolio.

To address obstructions located beyond the installation boundary that pose a risk to flying operations or the mission in general, identify and brief them to the Installation Mission Sustainment Team (IMST). This group can then determine the risk utilizing a mission sustainment risk report (MSRR), and establish a course of action the installation should take to mitigate. Review AFI 90-2001, *Mission Sustainment Program*, for further information.

Siting Obstructions. Siting new obstructions should be avoided whenever possible. In circumstances where this is unavoidable, site in accordance with the Siting Playbook (In Draft) when programming for new construction projects. Special conditions including the requirement for temporary or permanent waivers pertains when siting flightline facilities, NAVAIDs, or other structures on or in close proximity to the airfield.

Project Programming. Obstructions identified and waived by temporary waivers must have projects programmed to mitigate, correct, or remove the violation. The AP&W Program Manager should contact the installation programmer to coordinate on airfield projects. The project number, action programmed to correct the violation, programmed FY and cost estimate should be provided in Item 4 of AF Form 505.

Obstruction Reduction

Installations should review all approved permanent waivers to evaluate whether they require modification or if they should change to temporary waivers and be programmed for future corrective action. Once a project to correct the violation has been completed, the associated permanent or temporary waiver can be initiated for removal during the installation's next scheduled Annual Airfield and Airspace Waiver Review. All waiver records (including but not limited to lists, maps, and so forth) should also be updated to reflect these changes. If an obstruction was mitigated but continues to present a risk to flying operations, the waiver should remain in place but amended to reference any changes in the size, location, or criteria violated by the obstruction.

Obstruction Management Checklist

This checklist facilitates the AP&W Program Manager in managing obstructions at the installation level.

☐ **Annual Airfield Assessment**

Annual assessment of imaginary airspace surfaces, airfield markings, and visual air navigational facilities to ensure the installation maintains compliance with standard airfield criteria.

☐ **Obstruction Analysis**

Analysis conducted once a potential obstruction has been identified. This includes verification of vertical obstruction violations using ArcGIS or other tools.

☐ **UFC Violations**

Reference UFC 3-260-01, *Airfield and Heliport Planning and Design*, UFC 3-260-04, *Airfield and Heliport Marking*, and UFC 3-535-01, *Visual Air Navigation Facilities* to verify whether potential obstructions violate airfield criteria.

☐ **Deliberate Risk Assessment (AF Form 4437)**

Completed AF Form 4437, *Risk Management (RM) Assessment*, signed by the Installation Commander. Use the form to evaluate the overall risk an obstruction presents to flying operations. Airfield Operations, the Base Civil Engineer (or alternate), and the Safety office will work together to complete AF Form 4437. The results from this form will help populate information in Item 5 of the *Airfield and Airspace Waiver Request Form*, AF Form 505.

☐ **FAA Notification (FAA OE/AAA Report) (If necessary)**

Copy of the approved FAA OE/AAA report. If necessary, Installations will file an evaluation request with the FAA using the Obstruction Evaluation/ Airport Airspace Analysis (OE/AAA) notice criteria tool. Submittal of an FAA Form 7460-1, Notice of Proposed Construction or Alteration, should be initiated approximately 45-days in advance of the scheduled construction activity to prevent delays to the project start date. In instances where there is a delayed response from the FAA, the Part 77 Case/ASN Number or filing date will suffice at the end of Item 5 on page 1 of the Airfield Waiver Request Form. https://oeaaa.faa.gov/oeaaa/external/portal.jsp

☐ **Course of Action**

☐ **Airfield and Airspace Waiver Request**

Prepare an Airfield and Airspace Waiver Request using *AF Form 505*. Reference the *Request for Waiver to Airfield and Airspace Criteria Instructions* section for guidance on completing and coordinating waiver request packages at the installation level.

- OR -

☐ **Exemption or Permissible Deviation**

Record exemptions or permissible deviations in accordance with UFC 3-260-01, Sections B13-2. Include these in the Airfield Waiver List (Tab 1) of the Annual Airfield and Airspace Waiver Review package.

☐ **NOTAMs**

Airfield Management shall file notices to Airmen (NOTAMs) to alert aircraft pilots of potential risks to flying operations.

☐ **Project Programming**

Project programmed to mitigate, correct or remove violations tied to temporary waivers.

☐ **Obstruction Reduction**

Corrective action to mitigate or remove the violation in question. Remove all associated airfield waivers once the project has corrected the violation.

Request for Waiver to Airfield and Airspace Criteria Instructions

What Is the Purpose of the Airfield and Airspace Waiver Request Form?

The *Airfield and Airspace Waiver Request Form*, AF Form 505, is used by the Base Community Planner (or alternate), Airfield Managers, and Safety personnel to submit waiver requests for obstructions to airfield and airspace criteria as defined by UFC guidelines. Memorandums are no longer required or permitted to request waivers to airfield and airspace criteria. Waivers are not required for exemptions or permissible deviations.

Per UFC 3-260-01, Para. B1-2.1.1, the action officer (AO) as appointed by the BCE, initiates an airfield waiver request when compliance with airfield criteria cannot be achieved, all alternatives are exhausted, and the obstruction is determined to pose little or no risk to flying operations. Submit all supporting documentation in accordance with the checklist on page 12.

AF Form 505 Instructions

Section I.

Waiver Number. Enter the assigned waiver number here using the standard numbering system below:

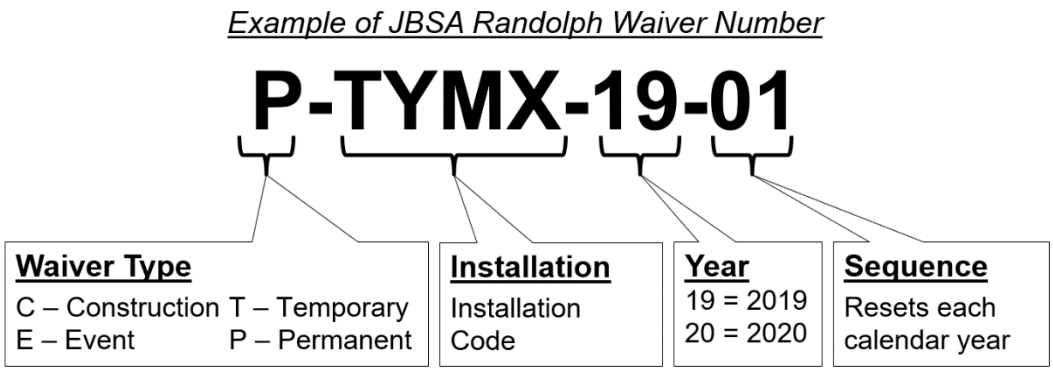

Figure 4 Example of a Waiver Number

The first letter indicates the type of waiver. The next four letters correspond to the Installation Code (TYMX – JBSA-Randolph), followed by two digits for the calendar year (CY) the waiver is initiated (18 – 2018). The last two digits denote the waiver sequence (i.e., 01, 02, 03…) for each waiver category. This number resets to '01' at the start of the new calendar year.

Proponent. Enter the requesting organization/symbol here.

Installation. Enter the installation name in the line provided (e.g., JBSA-Randolph AFB).

MAJCOM. Enter the responsible MAJCOM overseeing the installation.

Item Number 1. Action Requested. Select the option that best defines the waiver request:

Permanent Waiver – Where violations cannot be reasonably corrected.

Temporary Waiver – Where violations are to be corrected by a programmed project within the approved 8-year waiver validity period.

Construction Waiver – Where construction activities are scheduled to take place on the airfield and violate airfield criteria. Coordinate the waiver package at the installation level to ensure base leadership maintains situational awareness of the construction project. <u>These waivers must be finalized and signed, prior to the start of the construction project.</u>

Waiver Extension – Where additional time is required beyond the validity period granted by the original approved temporary waiver to complete a programmed project.

Waiver Amendment – Where significant changes are made to the original approved waiver.

Event Waiver – Where events of a temporary nature violate airfield criteria. These waivers are only valid for the duration of the event. *Air Show Waivers* are filed as event waivers.

Item Number 2. Criteria to be Waived. Select all applicable UFC criteria to be waived. Include the table number, paragraph, category, item, etc. for each violation excerpt as referenced in Tab 3 (e.g., Table 3-7, Item 7; Table 3-8, Item 3). In instances where multiple criteria are waived, note the UFC number at the beginning of each excerpt list. Begin with the most restrictive criteria and group violations to multiple criteria where practical.

Item Number 3. Description of Waiver. Clearly describe all violations to airfield criteria. List any airfield imaginary surfaces violated by the proposed action. Provide relevant details regarding elevations, heights, and distances affecting imaginary surfaces as supported by attached site location map (Tab 2).

For amendments to existing waivers, delineate changes at the end of the original text in Item 3 (e.g., Amendment 1: …, Amendment 2: …). Do not alter the original text of the waiver description.

For temporary airfield construction waivers, include pertinent details on the proposed construction activity, types and maximum height of construction equipment, laydown/storage areas, haul-routes, special access requirements to the airfield, and any other conditions unique to the project.

Item Number 4. Justification of Waiver. Provide an explanation to why airfield criteria needs to be waived. Temporary waiver requests must include the action planned to eliminate the violation. For construction waivers, provide the associated project number. For temporary waivers, provide the project number and programmed fiscal year (FY) associated with the corrective action when available. If a project number is not yet available, write "TBD" in this section.

Item Number 5. Risk Assessment Summary. Summarize key findings from the Risk Management (RM) Assessment (AF Form 4437) and provide the overall risk level with risk control implementation (i.e., low, medium, high, extremely high). Outline safety precautions, which will be taken to minimize hazards to flying operations, such as special painting, flags, or beacons to mark obstructions, briefing programs to flying personnel and/or construction crews on safety procedures for foreign object detection and control, or the issuance of Notices to Airmen (NOTAMs) by Airfield Management.

NOTE: Attach AF Form 4437 to the waiver package as Tab 3.

For temporary airfield construction waivers, indicate whether construction activities will occur during normal flying hours or equipment will be left on-site or taken off-site after use, as well as any foreign-object debris (FOD) concerns. Also note the displacement of any runway thresholds.

Provide the FAA OE/AAA Part 77 Case/ASN number if applicable in the line at the bottom of Item 5. Submittal of an FAA Form 7460-1, *Notice of Proposed Construction or Alteration* should be initiated approximately 45-days in advance of the scheduled construction activity to prevent delays to the project start date. Website: https://oeaaa.faa.gov/oeaaa/external/portal.jsp

Item Number 6. Installation Coordination and Approval. This section must be coordinated at the installation level regardless of the action requested. Coordination through these departments is required per UFC 3-260-01 Para. B1-2.2 and B1-2.3 for waiver requests as it streamlines the approval process and allows for greatest visibility by all functional offices. Terminal Instrument Procedures (TERPS) should sign in the block reserved for Airfield Ops (OSA)/TERPS. Airfield Ops should sign when the installation does not have a TERPS office. "Other" signature blocks are provided for instances where, Security Forces (SF), or Communications (CS) are suggested to review a waiver request.

> **NOTE:** Utilize your installations preferred process to coordinate the package. DO NOT modify signature blocks on the AF Form 505, with the exception of the "Other" boxes as required.

Enter the **period valid** corresponding to the type of waiver requested (i.e., "Indefinite" – permanent waivers, "8-Years" – temporary waivers, and "Duration of Construction" – construction waivers).

> **NOTE:** Installation/CC digital signature locks page 1 of the document.

After coordination and approval of AF Form 505 is accomplished at the installation level, submit the package to AFCEC/CP for further coordination. Send a copy of installation approved waivers to AFCEC/CP. Installations should maintain digital-copies of final approved waiver packages for record.

Section II. Coordination in Granting Waiver. To be completed by AFCEC (CP), AFIMSC (DET), and MAJCOM authority only. An "Other" signature block is provided for instances where Security Forces (A4S) or Communications (A6) are required to review a waiver request.

Section III. Waiver Approval. The responsible MAJCOM/CD approves or disapproves the waiver request once coordination/concurrence with all functional offices has been completed. The MAJCOM authorizing official may apply additional conditions when granting the waiver, if deemed necessary.

> **NOTE:** MAJCOM/CD digital signature locks the entire document (pages 1-2).

Upon MAJCOM approval, forward approved package to Installation/CC; Cc: DET, AFCEC/CP, Installation Airfield Manager (AM) and Action Officer (AO).

Permanent/Temporary Waiver Package Checklist

☐ **Tab 1. Airfield and Airspace Waiver Request Form (AF Form 505)**
Completed request form (Items 1-6, page 1), signed by the Installation/CC.

☐ **Tab 1.1. Airfield and Airspace Waiver Request Form (Original)**
Original approved waiver (required when requesting an amendment).

☐ **Tab 2. Site Location**
Site location graphics (i.e., maps, photos, etc.) depicting obstructions, imaginary surfaces to be violated, and area relative to the installation and airfield. For construction waivers include maps, photos, or graphics illustrating the proposed work location, laydown/storage areas, and haul routes.

☐ **Tab 2.1. Site Location (Original)**
Original site location graphics (required when requesting an amendment).

☐ **Tab 3. Deliberate Risk Assessment (AF Form 4437)**
Completed AF Form 4437 signed by the Installation/CC.

☐ **Tab 3.1. AF Form 4437 (Original)**
Original AF Form 4437 (only required when an amended waiver necessitates revisions to the prior risk assessment).

☐ **Tab 4. Violation Excerpts**
Highlighted UFC excerpts of violations to airfield criteria (i.e., table, item number, etc.). Do not attach the entire document, only pages containing excerpts relevant to the waiver.

☐ **Tab 5. Approved FAA OE/AAA Report** (if applicable)
Copy of the approved FAA OE/AAA report. In instances where there is a delayed (45+ days) response from the FAA, the Part 77 Case/ASN Number or filing date will suffice at the end of Item 5 on page 1 of AF Form 505.

☐ **Tab 0. Installation Coordination Sheet**
Digital/scanned copy of installation coordination sheet (wet or electronic signature is permitted).

References for Violations to Airfield Criteria

For *exemptions*, reference UFC 3-260-01, Section B1-2.1.1.

For *permissible deviations*, reference UFC 3-260-01, Section B13-2.

For *airfield markings*, reference UFC 3-260-04.

For *airfield lighting and NAVAIDS*, reference UFC 3-535-01.

Temporary Airfield Construction Waiver Package Checklist

☐ **Tab 1. Airfield and Airspace Waiver Request Form (AF Form 505)**
Completed request form (Items 1-6, page 1), signed by the Installation/CC prior to the start of the construction project.

☐ **Tab 2. Project Location**
Depiction of project location, including maps, photos, or graphics illustrating the proposed work location, laydown/storage areas, and haul routes.

☐ **Tab 3. Project Information**
Typically provided by the Project Manager, attach a copy of the Scope of Work (SOW) for the project and any other project information relevant to airfield safety concerns.

☐ **Tab 4. Deliberate Risk Assessment (AF Form 4437)**
Complete AF Form 4437 for Installation/CC signature.

☐ **Tab 5. Violation Excerpts**
Highlighted UFC excerpts of violations to airfield criteria (i.e., table, item number, etc.). Do not attach the entire document, only pages containing excerpts relevant to the waiver.

☐ **Tab 6. Approved FAA OE/AAA Report** (if applicable)
Copy of the approved FAA OE/AAA report. In instances where there is a delayed (45+ days) response from the FAA, the Part 77 Case/ASN Number or filing date will suffice at the end of Item 5 on page 1 of AF Form 505.

☐ **Tab 0. Installation Coordination Sheet**
Digital/scanned copy of installation coordination sheet (wet or electronic signature is permitted).

☐ **Send final copy to AFCEC AP&W Team**
Send a copy of the final, signed TACW to the AFCEC AP&W Team who will then forward it to MAJCOM for their SA. Send all packages to afcec.cp.workflow1@us.af.mil

Installation Annual Airfield and Airspace Waiver Review Process

To comply with UFC 3-260-01, *Airfield and Heliport Planning and Design*, Section B1-2.4., installations must conduct an annual review of all active approved permanent and temporary airfield waivers and their associated projects. Also, utilize this review period to identify exemptions and permissible deviations.

At a minimum, the Annual Airfield and Airspace Waiver Review meeting should include membership from Civil Engineering (CE), Airfield Management (AM), Airfield Operations/Terminal Procedures (TERPS), and Safety (SE). When necessary, representatives from Security Forces (SF), and Communications (CS) should be invited to provide relevant input on waivers. Airfield waivers are briefed quarterly at the Airfield Operations Board (AOB) per AFI 13-204V3, *Airfield Operations Procedures and Programs*, Section A3.2.6 & A3.3.8.

Conduct a review to assess whether waivers should remain active or be removed. Cancellation occurs in instances where waiver requirements are determined invalid or have been removed by a project. *Do not utilize the annual review to request new waivers*.

Validate if a requirement still exists to keep current waivers, permissible deviations, or exemptions. Ensure appropriate safety precautions and mitigation measures are being employed per AF Form 4437 and conditions of approval.

Facilities Board Validation. Present a summary of the Annual Airfield and Airspace Waiver Review to the Facilities Board for validation and document this action in the meeting minutes.

Package Submittal. Compile documentation from the annual waiver review meeting and Facilities Board. The package should contain the following items:

- Waiver List containing the number, type, description and validity for each approved waiver. For temporary waivers, provide any associated project numbers, estimated cost, programming and project status, and estimated completion date (ECD). Note whether waivers require cancellation. This document should also list all active exemptions and permissible deviations. Appendix B of this playbook is an example spreadsheet provided by AFCEC on the Airfield Planning & Waivers Toolbox.
- Documentation from the annual review meeting.
- Map displaying locations of all permanent and temporary waivers.
- Copy of the signed Facilities Board meeting minutes.

Per AFI 32-1015, para 2.32.6.2., submit a copy of the Annual Airfield and Airspace Review package to AFCEC/CP: AFCEC.CP.workflow1@us.af.mil.

AFCEC/CP will forward a copy of the package to the MAJCOM.

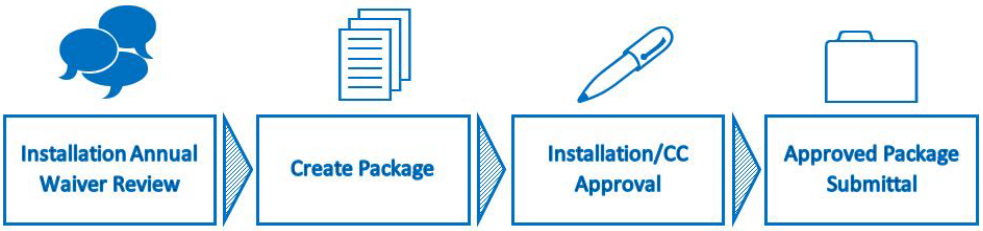

Figure 5 Waiver Review Process

Annual Airfield and Airspace Waiver Review Package Checklist

This checklist helps facilitate the Annual Airfield and Airspace Waiver Review process by ensuring all attachments are included in the package. You are not required to submit this checklist as part of the package.

☐ **Tab 1. Airfield Waiver List**
List of all active permanent/temporary airfield waivers, including any exemptions or permissible deviations.

☐ **Tab 2. Waiver Map**
Airfield map depicting locations for each approved permanent or temporary waiver, exemptions or permissible deviations, and established BRLs if applicable.

☐ **Tab 3. Annual Airfield and Airspace Meeting Minutes**
Signed meeting minutes from the installation's Annual Airfield and Airspace Waiver Review.

☐ **Tab 4. Annual Airfield and Airspace Review Meeting Slides** (if applicable)
Copy of the current year Annual Airfield and Airspace Review Meeting presentation.

☐ **Tab 5. Facilities Board Meeting Minutes**
Copy of the Facilities Board meeting minutes signed by the Installation/CC.

☐ **Tab 0. Installation Coordination Sheet**
Digital/scanned copy of installation coordination sheet (wet or electronic signature is permitted).

Glossary of References and Supporting Information

All AP&W forms can be found in the AFCEC Portal (SharePoint) Airfield Planning & Waivers Toolbox. The latest updated Air Force policy references are available via Air Force e Publishing and the UFC references are available on the Whole Building Design Guide website.

Please send all waiver requests packages, annual waiver review packages, or questions to AFCEC.CP.workflow1@us.af.mil.

Abbreviations and Acronyms

AC—Advisory Circular

ADCS—Approach Departure Clearance Surface

AF—Air Force

AFI—Air Force Instruction

AFIMSC—Air Force Installation and Mission Support Center

AFCEC—Air Force Civil Engineer Center

AGE—Aerospace Ground Equipment

AM—Airfield Management

AMU—Aircraft Maintenance Unit

AO—Action Officer

AORI—Airfield Obstruction Reduction Initiative

APZ—Accident Potential Zone

ASN—Aeronautical Study Number

BCE—Base Civil Engineer

BRL—Building Restriction Line

CC—Commander

CD—Deputy Commander

CP—Community Planner

CS—Communications Squadron

CZ—Clear Zone

DD—Decimal Degrees

DET—Detachment

ECD—Estimated Completion Date

eSSS—Electronic Staff Summary Sheet

FAA—Federal Aviation Administration

FOD—Foreign Object Debris

FY—Fiscal Year

GIS—Geographic Information Systems

GPS—Global Positioning System

IAW—In Accordance With

IMST—Installation Mission Sustainment Team

JBSA—Joint Base San Antonio

LG—Logistics

MAJCOM—Major Command

NAVAIDS—Navigational Aid System

NOTAM—Notice to Airmen

OE/AAA—Obstruction Evaluation / Airport Airspace Analysis

RM—Risk Management

SE—Safety

SFS—Security Forces Squadron

TERPS—Terminal Instrument Procedures

UFC—Unified Facilities Criteria

Forms

AF Form 505, *Airfield and Airspace Waiver Request Form*
AF Form 4437, *Deliberate Risk Assessment Form*
Form 7460-01, *Notice of Proposed Construction or Alteration*

References

AFI 13-204V3, *Airfield Operations Standardization and Evaluations,* 1 Sep 2010 (Change 2, 9 May 2019)
AFI 13-204V3, *Airfield Operations Procedures and Programs,* 1 Sep 2010 (Change 2, 29 Jun 2015)
AFI 32-1015, *Integrated Installation Planning,* 30 Jul 2019
AFI 90-2001, *Mission Sustainment Program,* 31 Jul 2019
FAA AC 150/5300-13A, *Airport Design*, 28 Sep 2012 (Change 1, 26 Feb 2014)
UFC 3-260-01, *Airfield and Heliport Planning and Design*, 4 Feb 2019
UFC 3-260-04, *Airfield and Heliport Marking*, 16 May 2018
UFC 3-535-01, *Visual Air Navigation Facilities*, 11 Apr 2017 (Change 1, 07, Mar 2018)
Federal Aviation Regulations, Part 77, *Objects Affecting Navigable Airspace*, 1 May 1965
Comprehensive Planning Playbook
Siting Playbook

Appendix A: Example Prefilled Permanent Waiver Request Form (Tab 1)

REQUEST FOR WAIVER TO AIRFIELD AND AIRSPACE CRITERIA

SECTION I. *(To be completed by installation)*

WAIVER NUMBER **P-ABCD-19-01**	PROPONENT **Requesting organization/symbol here.**
INSTALLATION **JBSA-Randolph AFB**	MAJCOM **Air Education Training Command (AETC)**

1. ACTION REQUESTED *(If submitting a request for a construction or Air Show waiver, complete items 1-6 only. Do not proceed to page 2.)*

PERMANENT WAIVER

☐ EXTENSION ☐ AMENDMENT
☐ AIR SHOW ☐ OTHER

2. CRITERIA TO BE WAIVED ☒ UFC 3-260-01 ☐ UFC 3-260-04 ☐ UFC 3-535-01 ☐ OTHER (If necessary you can select more than one regulation)

TABLE NUMBER, PARAGRAPH, CATEGORY, ITEM, ETC. *(e.g.,Table 3-7, Item 7; Table 3-8, Item 3)*

Section B13-2.21.2.8

3. DESCRIPTION OF VIOLATION *(Describe the obstruction and list any airfield imaginary surfaces to be violated. Support details with maps delineating imaginary surfaces. Provide pertinent elevations, heights, and distances impacting surfaces. For waivers to airfield marking or lighting, illustrate the change being requested.)*

(EXAMPLE) North and South End of Runway (EOR) buildings provide shelter to personnel who inspect and arm/disarm aircraft systems/components prior to takeoff/post recovery. The structures are within the graded portion of the Clear Zone and graded portion of the CZ and Primary Lateral clearance. All structures are outside of the mandatory frangibility zones and meet wingtip clearances, reference site plans attached. The facilities are small (26 ft 7 in x 26 ft, 7 in; 25 ft 2 in x 23 ft 6 in), single-story permanent structures (overall height less than 20 feet AGL) . The buildings violate the 50:1 Approach Departure Clearance Surface (ADCS); however, that was not a criteria when the facilities were constructed and initially approved for a permanent waiver. The shelters were originally constructed in 1977 as temporary shelters and then replaced with permanent construction in 1982 (reference documents included).

4. JUSTIFICATION OF WAIVER *(Explain why the criteria needs to be waived. Operational and mission concerns should be addressed. For temporary waiver requests, include action programmed to correct the violation, programmed FY, programmed cost estimate, and project number.)*

Per current guidance in UFC 3-260-01, B13-2.21.2.8, shelters are allowed, even within the graded portion of the Clear Zone, as long as they meet wingtip clearance requirements and do not violate the ADCS nor any runway/taxiway mandatory frangibility zones. The shelters meet all current requirements with the exception of the ADCS; reference graphics to see the locations of the respective facilities with respect to the ADCS.
BACKGROUND: A permanent waiver for the shelters was approved in 1977 (documentation included in this waiver request package). During the review of waivers in 2008, it was determined that a permanent waiver was not appropriate and that the facilities should be relocated to meet all current UFC 3-260-01 criteria. However, it is not viewed as being economically justifiable to alter/relocate the facilities given such a negligible improvement to safety of flight, and a permanent waiver is seen as the most correct course of action.
RECOMMENDATION: Given the current guidance, approval of a permanent waiver is requested so that the facilities may be retained for their intended purpose at their current locations.

PROJECT NUMBER *(Temporary & Construction Waivers Only)*

5. RISK ASSESSMENT SUMMARY *(Provide Risk Level and a brief summary of findings in the Risk Assessment Report. Attach AF Form 4437 to waiver package.)*

(Example) Risk assessment was conducted 24 Apr with airfield manager, SEF personnel, XX CES/CENP ,and base planner. Initial risk assessment was determined to be "medium." Assessment of Hazard Exposure: Based on the location of the facilities, which are offset by ~ 780 feet from the runway centerline, the exposure level component of the risk assessment is high, as aircraft taxi by and takeoff/land in close proximity when using runway.
Assessment of Hazard Severity: Determined to be Level 1, catastrophic, based on the most severe outcome potential of an aircraft striking the facilities in flight during takeoff/recovery. Assessment of Hazard Probability: Determined to be E, Unlikely, per the definition in AFPAM 90-803, page 25, paragraph 27.1.5.4: "All Airmen exposed - Occurs very rarely." This leads to an initial risk level of "Medium" per AFPAM 90-803, page 26, Figure 7. Control measures include aircrew training and education for awareness and use of obstruction lights. However, control measures do not meaningfully reduce the overall risk, and the residual risk level is "Medium." (see attached AF Form 4437)

FAA OE/AAA PART 77 CASE/ASN NUMBER

6. INSTALLATION COORDINATION AND APPROVAL *(Digital Signature)*

CIVIL ENGINEER (CE)	AIRFIELD MANAGER (AM)	AIRFIELD OPS (OSA) /TERPS	BASE SAFETY (SE)
OTHER	OTHER	OTHER	PERIOD VALID

INSTALLATION COMMANDER	SIGNATURE

AF505, 20190722 Prescribing Directive: AFI32-1015

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STOP! DO NOT CONTINUE FOR CONSTRUCTION OR AIR SHOW WAIVERS

REQUEST FOR WAIVER TO AIRFIELD AND AIRSPACE CRITERIA *(Continued)*		
SECTION II. COORDINATION IN GRANTING WAIVER *(To be completed by MAJCOM) (Digital Signature)*		
AFCEC (CP) *(Active Duty Only)*	AFIMSC (DET) *(Active Duty Only)*	LOGISTICS (A4)
OPERATIONS (A3)	SAFETY (SE)	OTHER
SECTION III. WAIVER APPROVAL *(To be completed by MAJCOM) (Digital Signature)*		
ADDITIONAL CONDITIONS APPLIED IN GRANTING WAIVER		
TITLE OF AUTHORIZING OFFICIAL	SIGNATURE	

Active Duty: Upon MAJCOM approval, forward package to Installation/CC; Cc: DET, AFCEC/CP, Installation AM and Action Officer.
Air National Guard: Upon MAJCOM approval, forward package to Installation BCE.
Refer to the *Airfield Planning Playbook* for specific guidance on filling out this form.

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Appendix B: Annual Airfield & Airspace Waiver Review Sheet

INSTALLATION NAME	Moody AFB	ANNUAL AIRFIELD & AIRSPACE WAIVER REVIEW	2019
MAJCOM	ACC		
INSTALLATION POC	Derrick Burke, derrick.burke@us.af.mil, DSN 460-9006		
INSTALLATION/CC APPROVAL DATE	Col Jennifer M. Short, Approved 12 May 19		

WAIVER (DO NOT INCLUDE CONSTRUCTION OR AIR SHOW WAIVERS)									PROJECT PROGRAMMING	
CURRENT WAIVER ID	NEW WAIVER ID	WAIVER DESCRIPTION	WAIVER TYPE	RISK LEVEL	DATE APPROVED	TERM	ORM REVIEW	REMOVE	CORRECTIVE ACTION	PROJECT STATUS
QSEU-001W		**Communications Antenna; B657;** violated primary lateral surface; above BRL	Permanent	n/a	28-Nov-1979	Indefinite	n/a	Y	Object has been removed	n/a
QSEU-002W		**Floodlight Pole, B617;** 140' pole violates 7:1 transitional surface	Permanent	Medium	18-Aug-1976	Indefinite	Y	N	QSEU18-0125	$1.41M, Programmed, FY21
QSEU-008W		**Aerospace Ground Parking Area (AGE);** violates primary surface	Permanent	n/a	21-Mar-1977	Indefinite	n/a	Y	Area no longer used for AGE	n/a
QSEU-014W		**Aircraft Parking on NW Ramp;** violates the primary lateral clearance and graded portion of the CZ	Permanent	Medium	9-May-1977	Indefinite	Y	N	n/a	n/a
QSEU-042W		**Clear Rinse & Support Facility;** violates graded area of the CZ	Permanent	Medium	17-Oct-2000	Indefinite	Y	N	n/a	n/a
QSEU-043W		**HH60 Hot Pit Refueling Area;** violates primary lateral clearance	Permanent	Medium	31-Jan-2007	Indefinite	Y	Y	Hot Pit refueling exempt per UFC 3-260-01, Table 3.2, Item 12	n/a
QSEU-045W		**Apron A, B, & Hot Cargo Pad Flood Lights;** HCP/Apron B violates 7:1; Apron A violates primary lateral surface	Permanent	Medium	9-May-2009	Indefinite	Y	N	QSEU14-9001 replaced HCP and Apron B lights/poles; no longer violate 7:1; QSEU18-0125 to address remaining lights	$3.6M, Completed, FY18 (QSEU14-9001) $1.41M, Programmed, FY21 (QSEU18-0125)
QSEU-047W		**Aerial Delivery Facility;** violates 7:1 transitional surface	Permanent	Medium	10-May-2009	Indefinite	Y	N	n/a	n/a
QSEU-050W		**Base Ops Marquee & Covered Walkway;** violates primary lateral surface	Permanent	Low	11-May-2009	Indefinite	Y	N	n/a	n/a
QSEU-051W		**ATC Transmitter Towers, B1501;** violate the 7:1 transitional surface	Permanent	Medium	18-Aug-1976	Indefinite	Y	N	n/a	n/a
QSEU-055W		**A-10 Sunshades;** violates taxiway clearance (Ch 5)	Permanent	Medium	6-May-2009	Indefinite	Y	Y	Misapplied UFC criteria; Sunshades meet wingtip clearance reqts. for taxilanes per Chap. 6 para 2-8.2.5	n/a
QSEU-009W		**Trees violating airfield surfaces;** violate the 7:1 transitional surface, graded area of the CZ and the ADCS	Temporary	Medium	6-May-2009	8 years	Y	N	QSEU06-0214 QSEU19-0130	$1.76M, Programmed, FY21 (QSEU06-0214) $80K, Programmed, FY21 (QSEU19-0130)
QSEU-048W		**Above Ground Electrical Boxes;** violates primary lateral surface and graded area of the CZ	Temporary	Medium	6-May-2009	8 years	Y	Y	Project QSEU005000 removed/relocated electrical boxes	$2.2M, Completed, FY19
QSEU-052W		**Burma Road at Mission Lake;** violates the graded portion of the CZ	Temporary	Medium	6-May-2009	8 Years	Y	N	QSEU06-0214	$1.76M, Programmed, FY21
QSEU-053W		**North End of Runway Building (EOR);** violate the ADCS	Temporary	Medium	6-May-2009	8 Years	Y	N	Requesting single permanent waiver in a separate package	n/a
QSEU-054W		**South End of Runway Building (EOR);** Facility violates ADCS	Temporary	Low	6-May-2009	8 Years	Y	N		
QSEU-056W		**Flight line Security Fence;** violates the graded portion of the CZ	Temporary	Low	6-May-2009	8 Years	Y	N	QSEU06-0214	$1.76M, Programmed, FY21

PLEASE SEND ANNUAL WAIVER REVIEW PACKAGES
OR QUESTIONS TO: AFCEC.CP.workflow1@us.af.mil

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INSTALLATION ANNUAL